MIL-S-52520A
23 September 1971
SUPERSEDING
MIL-S-52520
15 June 1966

MILITARY SPECIFICATION

SNOWPLOWS, VEHICLE MOUNTING; STRAIGHT BLADE,

REVERSIBLE, AND V-BLADE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers snowplows for mounting on commercial and military vehicles, including 2-1/2- to 5-ton trucks, 34,000 GVW carriers, 10,000 pound rough terrain fork lift trucks, graders and loaders.

1.2 Classification. Snowplows covered by this specification shall be of the following types and sizes as specified (see 6.3):

Type I - Straight blade reversible.
Size 1 - 24 inch high moldboard - 7 foot, 3 inch cut.
Size 2 - 30 inch high moldboard - 8 foot cut.
Size 3 - 36 inch high moldboard - 8 foot, 9 inch cut.

Type II - V-Blade.
Size 1 - 8 feet wide at cutting edge - nose height 2 feet 6 inches.
Size 2 - 8 feet wide at cutting edge - nose height 3 feet 5 inches.
Size 3 - 9 feet wide at cutting edge - nose height 4 feet 6 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.

FSC 3830

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the currentversion before use.

PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Military

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-H-14435	- Hitches, Snowplow, Front-of-Vehicle Mounting.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A327 - Impact Testing of Cast Iron.
A570 - Hot-Rolled Carbon Steel Sheets and Strip, Structural Quality.
E10 - Brinell Hardness of Metallic Materials.
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103).

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure, B3.0.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, *Section IX, Welding Qualifications.*

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The snowplows shall be the supplier's standard commercial models, except as specified herein and shall be of the following types:

(a) Straight blade reversible, trip-over with automatic recovery.
(b) V-blade.

3.1.1 Mounting. The reversible and the V-blade type snowplows shall be mountable and operate on a hitch conforming to MIL-H-14435, type II.

3.2 First article (preproduction model). The supplier shall furnish one snowplow of each type and size as applicable, for examination and testing within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce snowplows that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Initial production. When specified, the supplier shall furnish to the Government one or more snowplows, for inspection as specified in 4.4.

3.4 Material. Material shall be steel, or aluminum as specified herein. Snowplow moldboards shall be fabricated from ASTM A570, Grade A, B, C, D, or E as required, sheet steel not thinner than 10 gage U. S. Standard. When specified, the moldboard shall be fabricated from SAE AA 5086-H32 aluminum and the structural members shall be fabricated from SAE AA 6061-T6 aluminum. Cutting edges for the snowplows shall be not less than 1/2 inch thick by 8 inches wide, and shall be of high-carbon heat-treated steel, SAE 1080 or SAE 1090, with a Brinell hardness of 250 to 325. Materials not specified shall be those used for the same purpose by the supplier on the standard snowplow, of the size specified, that he regularly sells on the commercial market and shall be subject to all provisions of this specification. Material shall be new.

3.4.1 Threaded parts. Threaded parts shall be in accordance with Handbook H28.

3.5 Ease of maintenance. All major assemblies shall be accessible for servicing, repair, and replacement.

3.6 Ground clearance. The ground clearance of the mounted snowplows when raised for transport shall be not less than 8 inches.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

3.7 Use. The snowplows shall withstand without permanent deformation, damage or interference with the carrier tires, plowing operations at speeds up to 20 mph on improved and unimproved roads, or when specified, at speeds up to 35 mph.

3.8 Lubrication. All surfaces of the snowplow assembly requiring lubrication shall be provided with means for lubricating.

3.9 Identification marking. The snowplows shall be identified in accordance with MIL-STD-130. The marking shall be applied to the snowplows on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.9.1 Shipping data plate. A shipping data plate(s) conforming to MIL-P-514, type III, composition C of type I, grade A, class 1 material shall be provided. In addition, the plate(s) shall show the silhouette of the snowplows in transport position indicating the center of gravity and the location of the lifting and tiedown points. The plate(s) shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.10 Instruction plates. Each snowplow shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, and servicing the snowplow. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.11 Treatment and painting. The portions of the snowplows normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. Final paint coat color shall be full gloss yellow No. 13538. Where dissimilar metals are used, the contact surfaces shall be treated and painted in accordance with MIL-T-704.

3.12 Instructional material. Two copies of an illustrated list of all parts and two copies of the supplier's maintenance service and operation manual shall be included with each snowplow.

3.13 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.14 Items to be furnished with snowplows. No shipment of snowplows shall be made unless repair parts, tools, instructional material, and accessories are included with the shipment or unless approval for shipment of the snowplows without such items has been received from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

3.15 Physical characteristics. Physical characteristics of the snowplows shall be as specified in table I or table II as applicable.

Table I. Straight Blade Reversible

Size	1	2	3
Moldboard height without deflector, nominal.	24 in.	30 in.	36 in.
Shoes under cutting edge, minimum.	2	2	2
Width of path cleared at 37 degrees, nominal.	7 ft. 3 in.	8 ft.	8 ft. 9 in.
Bulldozing, nominal.	9 ft.	10 ft.	11 ft.

Table II. V-Blade

Size	1	2	3
Width at cutting edge, nominal.	8 ft.	8 ft.	8 ft.
Moldboard, spread, nominal.	10 ft. 2 in.	10 ft. 10 in.	11 ft.
Height at nose, nominal.	2 ft. 6 in.	3 ft. 5 in.	4 ft. 6 in.
Height at rear of moldboard, nominal.	5 ft. 2 in.	6 ft. 4 in.	7 ft. 6 in.
Shoes under cutting edge, minimum.	3	3	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

3.16 Type I - reversible blade. Reversible blade snowplows shall conform to table I and as specified herein.

3.16.1 Reversing mechanism. The moldboard shall be manually reversible to the right and to the left. Positioning of the snowplow to the right and to the left shall be accomplished by one man without the use of tools. A positive lock shall be provided to enable the operator to maintain the moldboard at various angles including, but not necessarily limited to angles of 0 degrees, 28 degrees, 35 degrees, and 42 degrees to the right and to the left when measured perpendicular to the longitudinal axis of the vehicle chassis. When specified, the reversing mechanism shall be hydraulically powered. The hydraulic power shall be provided by the pump used to furnish power to the hitch.

3.16.1.1 Controls. All controls for the hydraulic operation of the snowplows shall be furnished and mounted in the operator's compartment of the snowplow carrier, within reach of the operator in his seated operating position.

3.16.2 Moldboard. The moldboard shall be paneled or rolled and shall have reinforcing ribs to provide a rigid structure. The moldboard shall be electrically welded to a moldboard framework. The bottom edge of the moldboard framework shall be steel or aluminum (see 3.4), angle not less than 5 inches by 3-1/2 inches by 1/2 inch and shall serve as a reinforcement for the cutting edge. When the moldboard is fabricated from aluminum, all aluminum-to-steel connection points on the structural members shall be reinforced with wear plates, and attached to the framework with removable pins or bolts.

3.16.3 Moldboard deflector. When specified, the moldboard shall be equipped with a full-length, hinged-type deflector equipped with not less than four spring-loaded hinges, or when specified shall be hydraulically operated and controlled. The deflector shall be made from the same material as the moldboard and shall have a box brake of the same material at the outer edge to provide a rigid structure.

3.16.4 Shoes. Four shoes of chill cast alloy white iron shall be provided. One shoe shall be bolted near the right end of the moldboard framework and one shoe bolted near the left end. Two curb shoes shall be bolted on the outer edges of the moldboard assembly. All shoes shall have a wearing surface hardness of Rockwell C40 to C60 and shall have an Izod impact value of 20-35 foot-pounds at minus 10° F.

3.16.5 Framework adjustment and travel lock. The snowplow framework shall be of steel and shall be adjustable to the following positions by one man without the use of tools:

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

(a) The weight of the moldboard and part of the drive frame is supported by the shoes.
(b) The major portion of the weight of the plow is applied at the cutting edge.
(c) The plow is lifted free of the ground and in a positive mechanical lock for travel.

3.16.6 Shock-absorbing and trip-over unit. The snowplow framework shall be equipped with a shock-absorbing and trip-over unit. The unit shall minimize road shocks and shall trip the moldboard without permanent deformation or damage which impairs the required operational and functional performance when immovable obstacles up to 6 inches high are encountered by the moldboard. The unit shall automatically return the moldboard to its correct plowing position after passing the obstacle.

3.16.7 Cutting edge(s). The cutting edge(s) shall extend not less than the length of the moldboard. The cutting edge shall be of high-carbon-alloy, heat-treated steel, SAE 1080 or SAE 1090 with a Brinell hardness of 250 to 325. The cutting edge(s) shall be bolted to the moldboard and shall be replaceable and reversible, edge for edge and end for end.

3.17 Type II - V-blade. V-blade snowplows shall conform to table II and as specified herein.

3.17.1 Moldboard. The moldboard shall be paneled or rolled, of steel or aluminum as specified (see 3.4), and shall be provided with a reinforced box brake along the top edge.

3.17.1.1 Moldboard deflectors. When specified, the moldboard shall be equipped with full length, hinged-type deflectors, each furnished with not less than four spring-loaded hinges, or when specified, shall be hydraulically operated and controlled. The deflectors shall be made of the same material as the moldboard and shall have a box brake of the same material at the outer edge to provide a rigid structure.

3.17.2 Cutting edges. The cutting edges shall extend approximately the same length as the bottom edges of the moldboard. Cutting edge material shall be of high-carbon alloy, heat-treated, abrasion-resistant steel, SAE 1080 or SAE 1090 with a Brinell hardness of 250 to 325. Cutting edges shall be bolted to the moldboard and shall be reversible, edge for edge and end for end.

3.17.3 Plow shoes. Plow shoes shall be bolted to the plow frame and shall be replaceable.

3.17.3.1 Nose shoe. One nose shoe shall form the central leading point of the vee shape formed by the cutting edges. The nose shoe shall be of heat-treated, abrasion-resistant steel.

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:21Z
Check the source to verify that this is the current version before use.

3.17.3.2 Weight bearing shoes. Two weight-bearing shoes made of chill cast alloy white iron shall be installed on the plow frame. The shoes shall have a wearing surface hardness of Rockwell C40 to C60 and shall have an Izod impact value of 20 to 35 foot-pounds at minus 10°F.

3.17.4 Plow framework. The framework for the plow shall be made of steel or aluminum members as specified (see 3.4). Structural members subject to bending or other damage shall be bolted together. All other members shall have hot-riveted or electric-welded connections.

3.17.5 Drive bars. Drive bars shall be pin connected at both ends. For the size 1 snowplow, the snowplow drive bars shall be of the tubular, shock-absorbing type, not less than 2-1/2-inch diameter extra heavy duty pipe with a solid insert of not less than 2-1/8-inch diameter. Shock springs shall be of not less than 5/8-inch diameter spring steel wire, 3-1/2-inch outside diameter, and 8-inch free length. Drive bars shall be of not less than 4 by 4 by 1/2-inch steel or aluminum angles for size 2 and 3 snowplows.

3.17.6 Cutting edge adjustment. The height of the cutting edge above the road surface shall be adjustable without the use of tools.

3.18 Government-loaned property. The following property as specified, in the quantity indicated will be loaned by the Government (see 6.5):

Table III - Government-loaned Property

Item No.	Description	Identification	Quantity
1.	Hitch, Snowplow, Front-of-Vehicle Mounting	FSN 3830-010-4939 (for 5-ton military truck, M39 series)	1
	or		
2.	Hitch, Snowplow, Front-of-Vehicle Mounting	FSN 3830-010-4940 (for 10,000 lb. rough terrain fork lift truck)	1
	or		
3.	Hitch, Snowplow, Front-of Vehicle Mounting	MIL-H-14435, type II	1
	and		
4.	Truck, 5-ton, 6 by 6, M39 Series	MIL-T-740	1

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the current version before use.

Table III - Government-loaned Property (continued)

Item No.	Description	Identification	Quantity
	or		
5.	Truck, Forklift, Rough Terrain, 10,000 lbs. MIL-T-21864	FSN 3930-073-8750	1
	or		
6.	Truck, commercial or military, (other than as listed)	Make and model as specified.	1
	or		
7.	Grader, road, motorized	Make and model unspecified.	1
	or		
8.	Loader, scoop-type pneumatic-tired.	Make and model unspecified.	1

3.19 Workmanship. The snowplows shall be free from defects such as incomplete welds, rust, cracks and other defects that could impair their operation or serviceability.

3.19.1 Metal fabrication. Metal used in the fabrication of the snowplow shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of metal, may be employed instead of shearing or sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated metal shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges and corners are required.

3.19.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be correctly torqued.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the currenversion before use.

3.19.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.19.4 Welders and welding.

3.19.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

3.19.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the weld or of the part. All weldments shall be free from slag, flux, weld spatter, and other impurities.

3.19.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein.

3.19.6 Aluminum fabrication. Fabrication of aluminum shall be done by methods that will not cause injury to the metal. Care shall be used in handling to avoid scratching and damage to the edges. Finished edges shall be smooth and free from lamination, kinks, bends, winds, and scratches. Corners shall be square and true. Bends shall be made with metal dies or fixtures to assure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the currentversion before use.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection. Prior to examination and test of the snowplow, the following operations shall be performed:

(a) Service the snowplow with grease (and oils if applicable) designated for use in the ambient temperatures at which the test will be run.
(b) Adjust spring-loaded mechanisms, and other components requiring adjustment, to their required settings.

4.3.1 Examination. Prior to testing, the snowplow shall be examined for defects marked "X" in column 1 of table IV. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the snowplow shall be subjected to the tests marked "X" in column 2 of table V. Failure of any test shall be cause for rejection. The sequence of testing shall be at the option of the supplier.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production snowplows will be selected at random by the Government from snowplows being produced from production tooling and will be examined for the defects marked "X" in column 2 of table IV and subjected to the applicable tests marked "X" in column 3 of table V to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production snowplow shall not exclude the remaining snowplows from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of an initial production snowplow to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production snowplows(s) and shall be cause for refusal by the Government to continue

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the current version before use.

acceptance of production snowplows until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on snowplows previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all snowplows accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each snowplow shall be examined for the defects marked "X" in column 3 of table IV. Presence of one or more defects shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table IV.

Table IV. Examination Schedule

Preproduction	Production		Defect	Requirement Paragraph
Pretest	Initial	Individual		
1	2	3	4	5
X	X	-	101. Materials not as specified.	3.4
X	X	-	102. Major assemblies not accessible for servicing and repair.	3.5
X	X	X	103. Dimensions not as specified.	3.15
X	X	-	104. Reversing mechanism not as specified for type I snowplow.	3.16.1
X	X	-	105. Moldboard not as specified.	3.16.2, 3.17.1

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

Table IV. Examination Schedule (continued)

1	2	3	4	5
X	X	-	106. Moldboard deflector (when required) not as specified.	3.16.3, 3.17.1.1
X	X	-	107. Shoes on moldboard and pushframe not as specified.	3.16.4, 3.17.3
X	X	-	108. Framework adjustment and travel lock not as specified for type I snowplows.	3.16.5
X	X	-	109. Shock-absorbing and tripover not as specified for type I snowplows.	3.16.6
X	X	-	110. Cutting edge(s) not as specified.	3.16.7, 3.17.2
X	X	-	111. Drive bar type and dimensions not as specified for type II snowplows.	3.17.5
X	X	-	112. Cutting edge height not adjustable for type II snowplows.	3.17.6
X	X	X	113. Lubricating means not as specified.	3.8
X	X	X	114. Identification marking not as specified.	3.9
X	X	X	115. Treatment and painting not as specified.	3.11
-	X	X	116. Items furnished with snowplow not as specified.	3.14
X	X	X	117. Workmanship not as specified.	3.19
X	X	X	118. Assembly not correct.	Section 3
X	X	X	119. Instruction plate not as specified.	3.10.1
X	X	X	120. Repair parts and tools not furnished as specified.	3.13
X	-	-	121. Controls not where specified.	3.16.1.1
X	X	X	122. Instructional material not as specified.	3.12

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the current version before use.

4.6.2 Tests. Tests shall be conducted in accordance with the test schedule in table V.

Table V. Test Schedule

Item No.	Pre-Prod.	Init. Prod.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
1	X	X	Mounting and travel, type I	4.6.2.1	3.1.1, 3.6
2	X	-	Performance, type I	4.6.2.2	3.7, 3.16.6
3	X	X	Mounting and travel, type II	4.6.2.3	3.1.1, 3.6
4	X	-	Performance, type II	4.6.2.4	3.7
5	X	X	Impact test, snowplow shoe and material	4.6.2.5	3.16.4, 3.17.3.2
6	X	X	Hardness test	4.6.2.6	3.16.4, 3.16.7, 3.17.2, 3.17.3.2

4.6.2.1 Mounting and travel, type I, snowplows. Test the snowplow in accordance with Test Method No. 1. Nonconformance to 3.1.1 or 3.6, evidence of breakage or permanent deformation of any component, inability to raise, lower, and hold the snowplow, or interference with the carrier tires while steering shall constitute failure of this test.

4.6.2.2 Performance, type I, snowplows. Test the snowplow in accordance with Test Method No. 2. Nonconformance to 3.7 or 3.16.6 shall constitute failure of this test.

4.6.2.3 Mounting and travel, type II, snowplows. Test the snowplow in accordance with Test Method No. 3. Nonconformance to 3.1.1 or 3.6, evidence of breakage or permanent deformation of any component, inability to raise, lower, and hold the snowplow, or interference with the carrier tires while steering shall constitute failure of this test.

4.6.2.4 Performance, type II, snowplows. Test the snowplow in accordance with Test Method No. 4. Nonconformance to 3.7 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

4.6.2.5 Impact, snowplow shoe material. Determine the Izod impact value of a specimen of the plow shoe material at minus 10° F., in accordance with Test Method No. 5. Nonconformance to 3.16.4 or 3.17.3.2, as applicable, shall constitute failure of this test.

4.6.2.6 Hardness. Test the cutting edge and shoes for hardness in accordance with Test Method No. 6. Nonconformance to 3.16.7 or 3.17.2 as applicable, shall constitute failure of this test.

4.7 Inspection comparison. When initial production inspection is specified, the Government may select snowplows at any time during the contract production period and subject these snowplows to the examination specified in table IV and the applicable tests specified in table V to determine that the quality of the selected snowplow is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Snowplows will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production snowplows.

4.7.1 Inspection failure. Failure of any snowplow when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of snowplows until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of crated snowplows for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

123. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
124. Surfaces requiring a contact preservative in accordance with the referenced document not coated with preservative for level A or B.
125. Hydraulic system not preserved as specified for level A or B.
126. Consolidated packaging not as specified for level A or B.
127. Blocking, bracing, anchoring, and waterproofing not in accordance with the referenced document for level A.
128. Strapping not zinc coated for level A.
129. Marking missing, illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified to prove prior to starting production packaging that the applied preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6).

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified.

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be in accordance with the supplier's recommendations. Removed bolts, pins, washers, and nuts shall be reinstalled in mating parts to prevent loss, or consolidated packaged as specified in 5.2.1.7.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with, MIL-P-116.

5.2.1.3 Unprotected metal surfaces. Unpainted exterior metal surfaces not specified herein, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

5.2.1.4 Hydraulic system. The hydraulic system shall be filled with the hydraulic fluid recommended by the supplier for operation, and the pump actuated to assure coating of all interior parts and surfaces. The pistons shall be fully retracted into the cylinders and secured. Any exposed portions of the piston rods shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with tape conforming to PPP-T-60, type IV. Any other exterior metal surfaces of components of the system requiring the application of a contact preservative shall be coated with type P-1 preservative. The control valve assembly and the hoses shall not be disconnected or the system drained unless recommended by the supplier. The control valve assembly shall be secured to the snowplow in a protected location or secured to the base of the crate specified in 5.3.1. When the control valve assembly or hoses are disconnected to accomplish the disassembly recommended by the supplier, all openings into the system shall be sealed with plastic or metal caps or plugs as applicable, and the control valve assembly packaged in a close-fitting box conforming to PPP-B-636, W5c, style optional. The contents shall be cushioned within the box to prevent movement.

5.2.1.5 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.7 Consolidated packaging. Repair parts, maintenance tools, technical publications, control valve assembly if removed, and any loose components removed in accordance with the supplier's recommendation shall be consolidated packaged, to prevent pilferage and loss, in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall not be required.

5.2.2 Level B. Each complete snowplow shall be preserved and packaged as specified in 5.2.1 for level A except fiberboard boxes used for packaging of the control valve assembly or hoses shall be type CF, class domestic, variety SW, grade as applicable to the weight of contents and the wood boxes for consolidated packaging shall be domestic type.

5.2.3 Level C. Each complete snowplow with repair parts, maintenance tools, and technical publications shall be preserved and packaged to afford protection against deterioration and damage from the supply source to the initial destination. The supplier's standard practice is acceptable when it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified.

5.3.1 Level A. Each complete snowplow, preserved and packaged as specified in 5.2 shall be packed in an open wood crate conforming to PPP-C-650, type V, style A, class 1, or in a open wood crate conforming to MIL-C-3774, to provide maximum compactness, low center of gravity, balance, and in a manner that void

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

spaces are held to a minimum. The contents shall be blocked, braced, anchored, and waterproofed with a shroud in accordance with MIL-STD-1186 and the appendix to the applicable crate specification. Crates shall be closed and strapped in accordance with the applicable specification. Strapping shall be zinc coated.

5.3.2 Level C. Each complete snowplow, preserved and packaged as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates. Containers, if used, shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The snowplows are intended for use in the removal of ice and snow from improved and unimproved roads, missile sites, R and U activities, helipads, and airfields.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of snowplow required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When initial production inspection is required and number of snowplows to be furnished, when applicable (see 3.3).
(e) When aluminum material is required (see 3.4).
(f) When plowing at speeds up to 35 mph is required (see 3.7).
(g) Applicable list of repair parts and maintenance tools (see 3.13).
(h) When a hydraulic reversing mechanism is required (see 3.16.1).
(i) When moldboard deflector(s) is required and whether deflector(s) will be equipped with spring-loaded hinges or be hydraulically controlled (see 3.16.3 and 3.17.1.1).
(j) Type of Government-loaned hitch and vehicle to be furnished (see 3.18).
(k) Time frame required for submission of preproduction pack (see 5.1).
(l) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when the snowplows will be held in covered storage.

6.3 Classification changes. Changes in classification of snowplows between this revision of the specification and the previous edition are as follows:

Old Designation	New Designation
Type I, Size 3-36 inch high moldboard-eleven foot cut.	- Type I, Size 3-36 inch high moldboard- 8 foot, 9 inch cut.
Type III-Rollover	- Deleted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

6.4 Preproduction model. Any changes or deviations of production snowplows from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish snowplows conforming to this specification.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.18.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the snowplows in accordance with this specification.

Custodians:

Army - ME
Air Force - 84

User interest:

Army - CE

Preparing activity:

Army - ME

Project No. 3830-0095

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:21Z
Check the source to verify that this is the currenversion before use.

APPENDIX

TEST METHOD NO. 1

Mounting and Travel, Type I

1. Test area: Level hardstand, with paved or improved surface; secondary gravel road.

2. Test equipment: Government-loaned snowplow carrier and hitch (see 3.18). Tools required for assembly. Measuring tape or ruler.

3. Procedure:

(a) Ballast the vehicle (when applicable) with rated load.
(b) Assemble the snowplow and hitch and mount on the carrier.
(c) Raise and lower the snowplow 25 times, from resting on the level surface to the full limit of lift each time. At the end of the 25 cycles and with plow in the full limit of lift, measure the ground clearance.
(d) With the snowplow at ground level plowing position, turn the carrier steering wheels to full right and full left, steer with the plow rotated to the maximum plowing angle to the right and to the left side. Check for clearance between the nearest part of the carrier and the snowplow in each of the four positions.
(e) Repeat (d) above, except with the plow raised to full lift in the carry position.
(f) With the snowplow in the raised position and held in this position by the hydraulic system only, drive the carrier for a distance for not less than 10 miles over secondary gravel roads at an average speed of 20 mph.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 2

Performance, Type I

1. Test course: Improved and unimproved roads covered with snow from 3 to 6 inches deep. Imbed a stone or metal object in the roadway so that it projects between 5 and 6 inches above the road surface, and shall not be dislodged or broken when struck by the snowplow. An existing immovable object may be used instead of the imbedded object.

2. Test equipment: Government-loaned snowplow carrier and hitch (see 3.18).

3. Procedure:

(a) Ballast the vehicle (when applicable) with rated load, and mount snowplow and hitch.
(b) Operate the carrier-mounted snowplow for not less than 2 hours at the specified speed (speeds up to 20 or 35 mph), as applicable to remove snow from the improved and unimproved roads.
(c) During the 2-hour test, drive the plow over the imbedded object not less than three times while traveling at not less than 20 mph so as to cause contact with the cutting edge of the moldboard.
(d) During the 2-hour test, position the manual-reversing plow at the available angled positions for 10 minutes or more in each position; or, when the power-reverse is required and furnished, change the plowing angle not less than 12 times while plowing, to include straight-ahead and maximum angle to each side.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T21:21Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 3

Mounting and Travel, Type II

1. Test area: Level hardstand, with paved or improved surface; secondary gravel road.

2. Test equipment: Government-loaned snowplow carrier and hitch (see 3.18). Tools required for assembly. Measuring tape or ruler.

3. Procedure:

(a) Ballast the vehicle (when applicable) with rated load.
(b) Assemble the snowplow and hitch and mount on the carrier.
(c) Raise and lower the snowplow 25 times, from resting on the level surface to the full limit of lift each time. At the end of the 25 cycles and with plow in the full limit of lift, measure the ground clearance.
(d) With the snowplow at ground level plowing position, turn the carrier steering wheels to full right and full left steer. Check for clearance between the nearest portion of the carrier and the snowplow in both steer positions.
(e) Repeat (d) above, except with the plow raised to full lift in the carry position.
(f) With the snowplow in the raised position and held in this position by the hydraulic system only, drive the carrier not less than 10 miles over secondary gravel roads at an average speed of 20 mph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 4

Performance, Type II

1. Test course: Improved and unimproved roads covered with snow from 8 to 15 inches deep.

2. Test equipment: Government-loaned snowplow carrier and hitch.

3. Procedure: Operate the carrier-mounted snowplow for not less than 2 hours at the specified speed (speeds up to 20 or 35 mph), as applicable to remove snow from the improved and unimproved roads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:21Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 5

Impact Test, Snowplow Shoe and Material

1. Test facility: The test facility shall consist of a laboratory or facility suitable for performing the test in accordance with ASTM A327.

2. Test apparatus: In accordance with ASTM A327.

3. Test procedure: Test the snowplow shoe in accordance with ASTM A327.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 6

Hardness Test

1. Test facility: The test facility shall consist of a laboratory or facility suitable for performing the test in accordance with ASTM E10 and ASTM E18.

2. Test apparatus: In accordance with ASTM E10 and ASTM E18.

3. Test procedure: Test the shoe and cutting edge material in accordance with ASTM E10 and ASTM E18.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: SNOWPLOWS, VEHICLE MOUNTING; STRAIGHT BLADE, REVERSIBLE, AND V-BLADE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:21Z
Check the source to verify that this is the current version before use.